

Machine Learning

Training Linear Regression- Week 5

Objectives

- Gradient Descent
- Reading Assignment (Andrew Ng , CS229 Lecture notes)

Slide credits:

Andrew Ng, CS229 Lecture notes

Where to find me?

Office:

Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM - 1:00 PM EST or by
appointment**

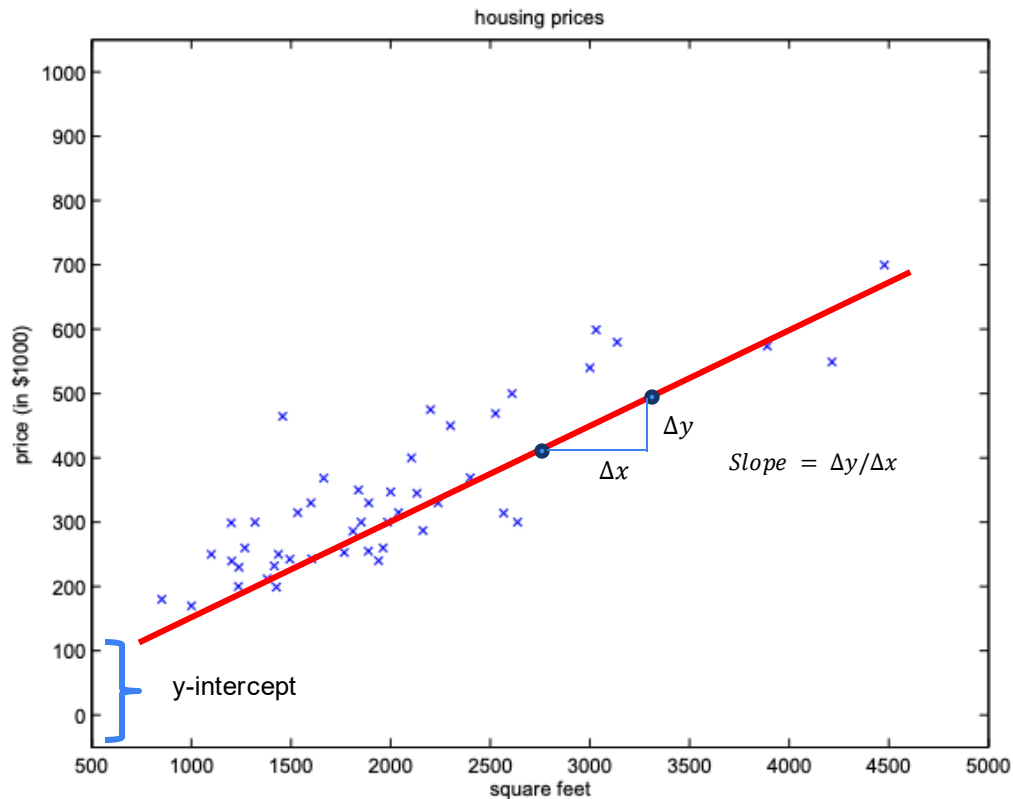
Linear Regression

The equation for linear regression,

$$f_{w,b}(x) = w * x + b$$

$w \rightarrow$ slope of the line

$b \rightarrow$ y-intercept



Cost function - Correction

For linear regression, the cost function, $J(w, b)$ usually measured as follows:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

or,

$$\frac{1}{2m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)})^2$$

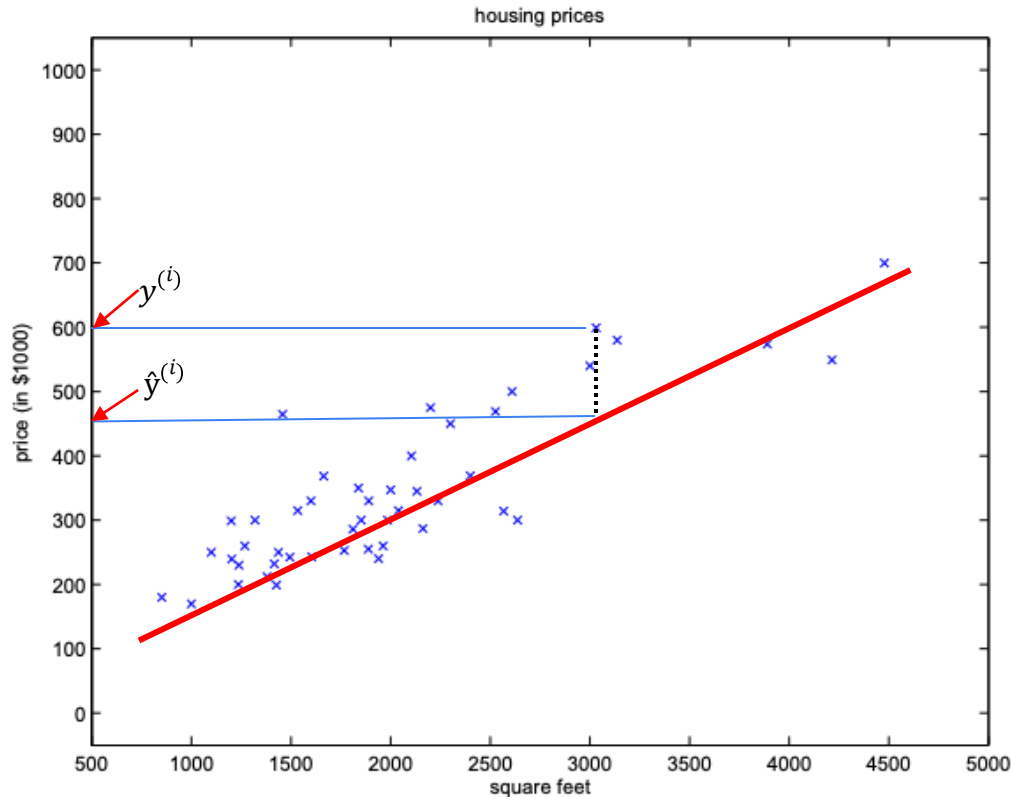
Substituting values for $f(x)$, we get:

$$\frac{1}{2m} \sum_{i=1}^m ((w \overset{\text{red } x}{x}^{(i)} + b) - y^{(i)})^2$$

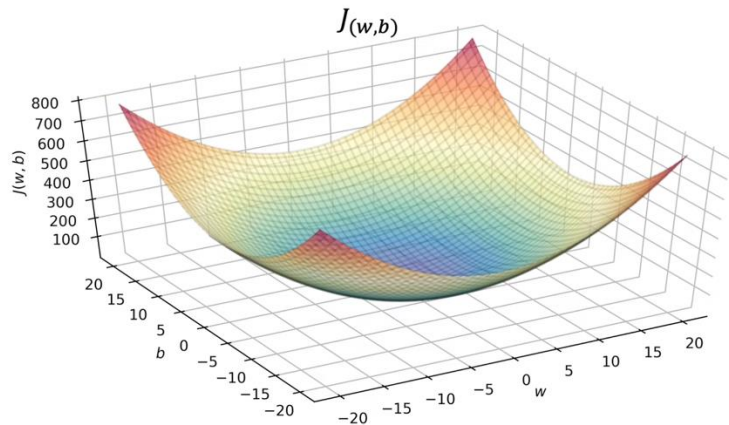
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$$\frac{1}{2m} \sum_{i=1}^m ((wx^{(i)} + b) - y^{(i)})^2$$

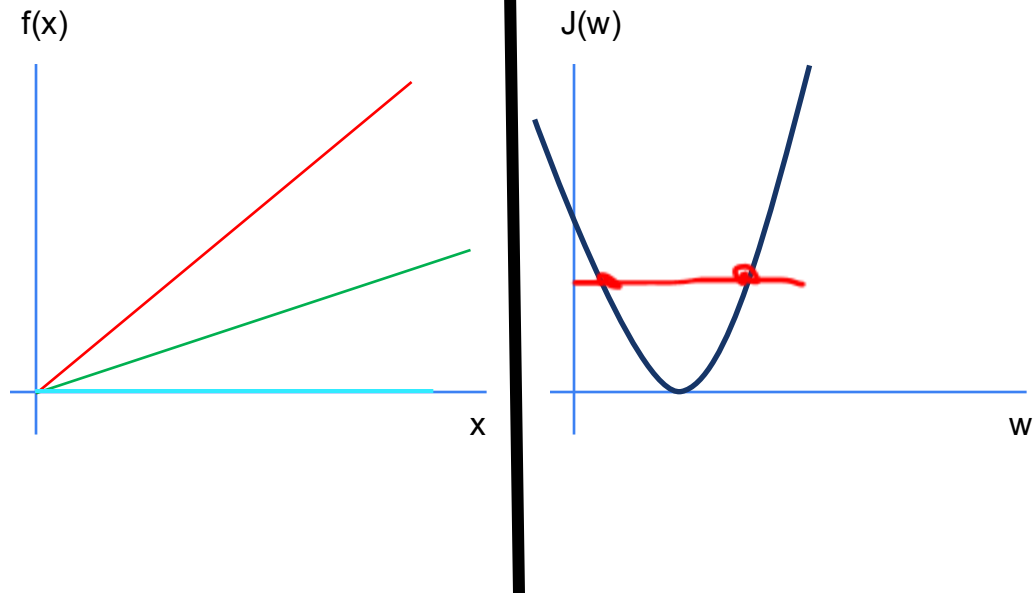
Our goal is to **minimize** $J(w, b)$



Cost Function Visualization

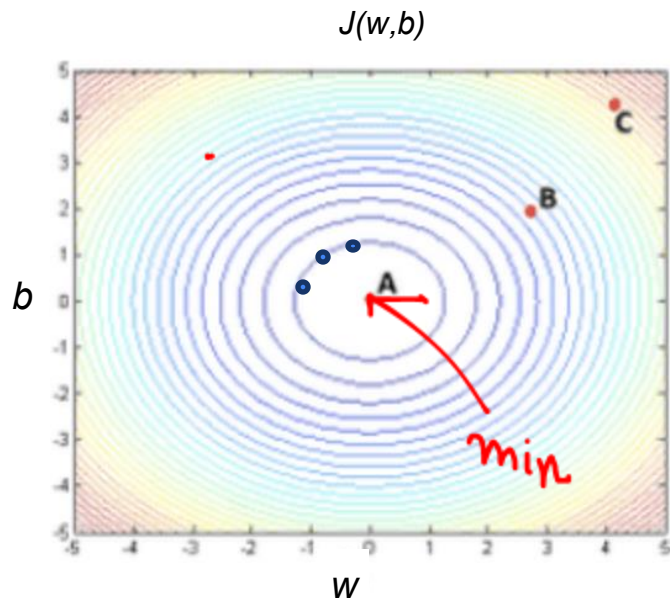


Plotting $J(w)$ for different values of w and b

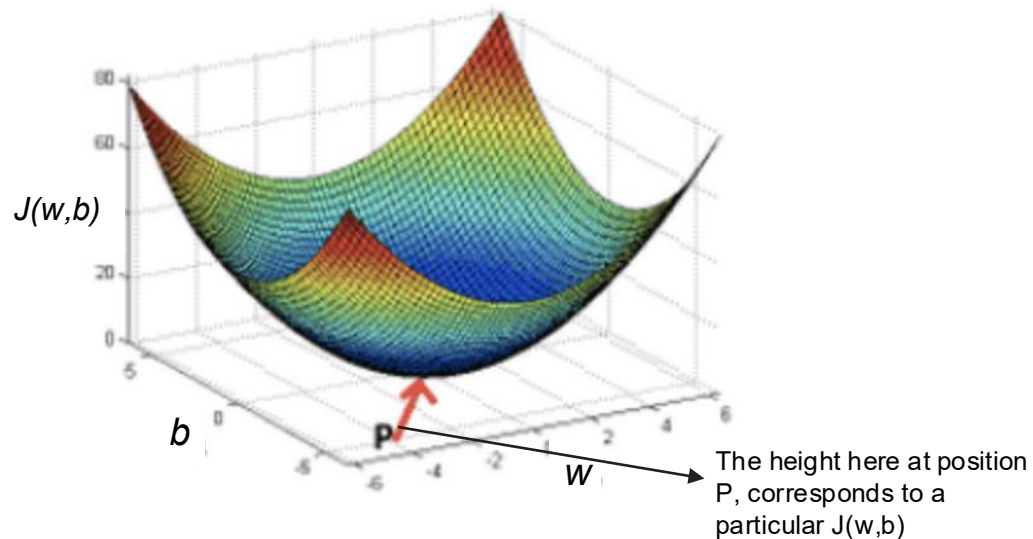


Plotting $f(x)$ and $J(w)$ for different values of x and w

Cost Function Visualization



Contour graph for $J(w)$



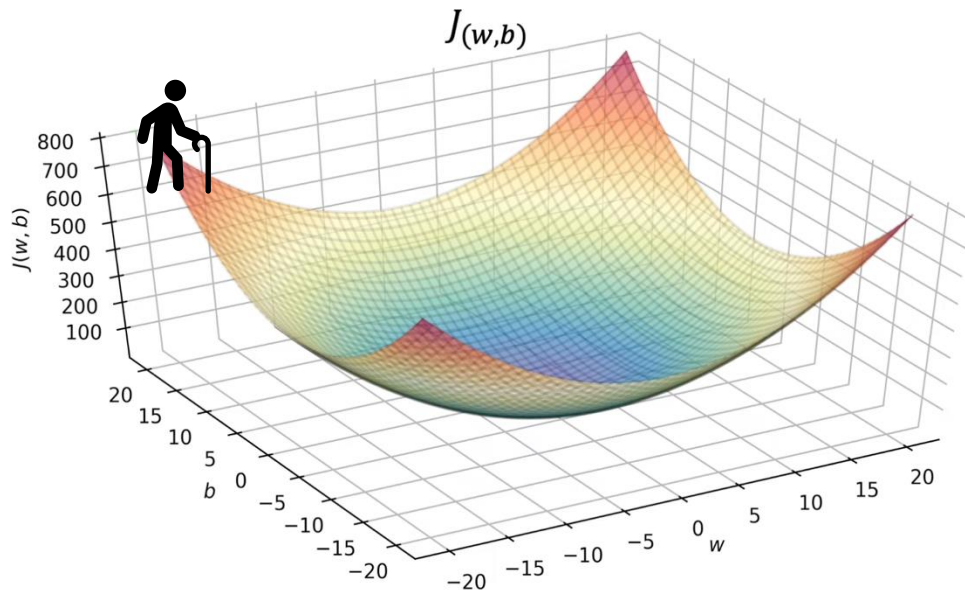
Plotting $J(w)$ for different values of w and b

Gradient Descent

The standard numerical way of minimizing $J(w, b)$ is gradient descent.

Intuitively, you can think of gradient descent as descending from a hill. We are assuming that you can't see too far, just your immediate surroundings.

You take a few steps in the direction that the hill is descending steepest. You repeat the process starting from the new point. Hopefully you will end up at the “bottom”.



Gradient Descent

The “bottom” here represents the location where $J(w, b)$ is **minimum**.

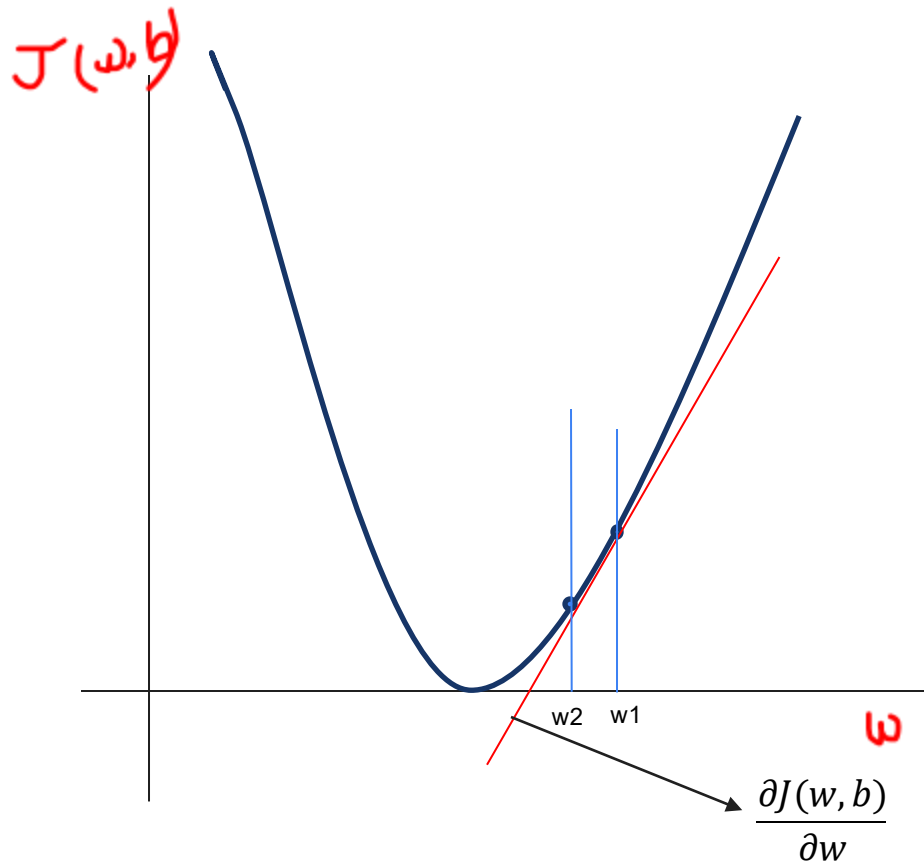
With gradient descent, we are trying to find values of w and b such that $J(w, b)$ is minimum.

Therefore, the **algorithm** can be written as:

Repeat until **convergence**:

$$w = w - \alpha \frac{\partial J(w, b)}{\partial w}$$

$$b = b - \alpha \frac{\partial J(w, b)}{\partial b}$$



Gradient Descent

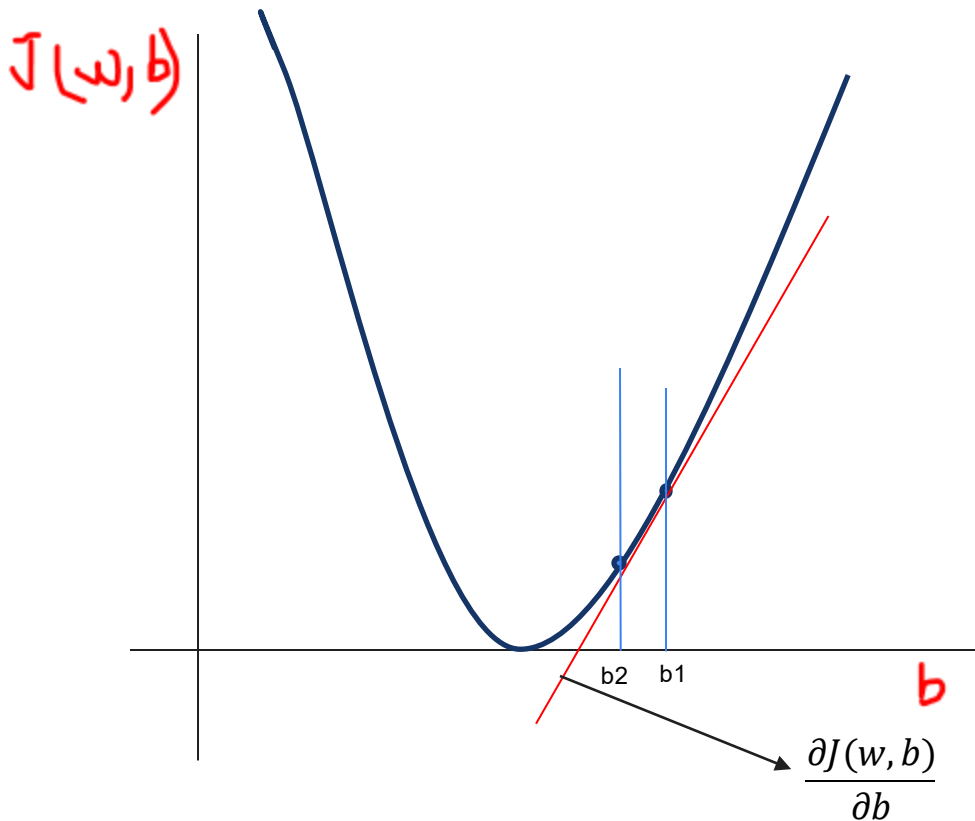
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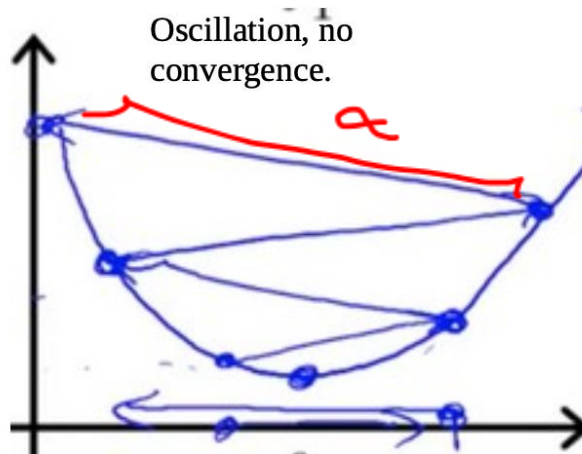
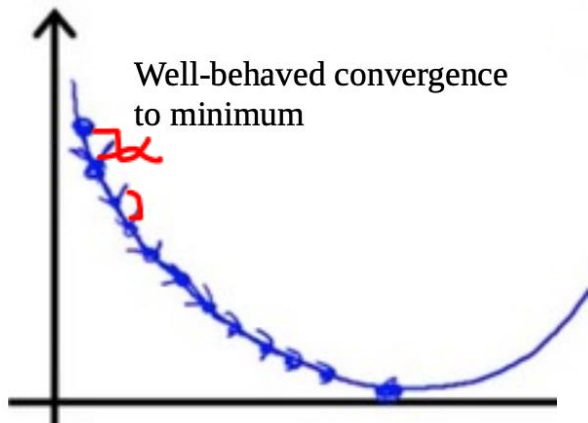
α is called learning rate. It is equivalent to the step one would take to descend the hill.

$\frac{\partial J(w,b)}{\partial w}$ and $\frac{\partial J(w,b)}{\partial b}$ provides the direction of movement



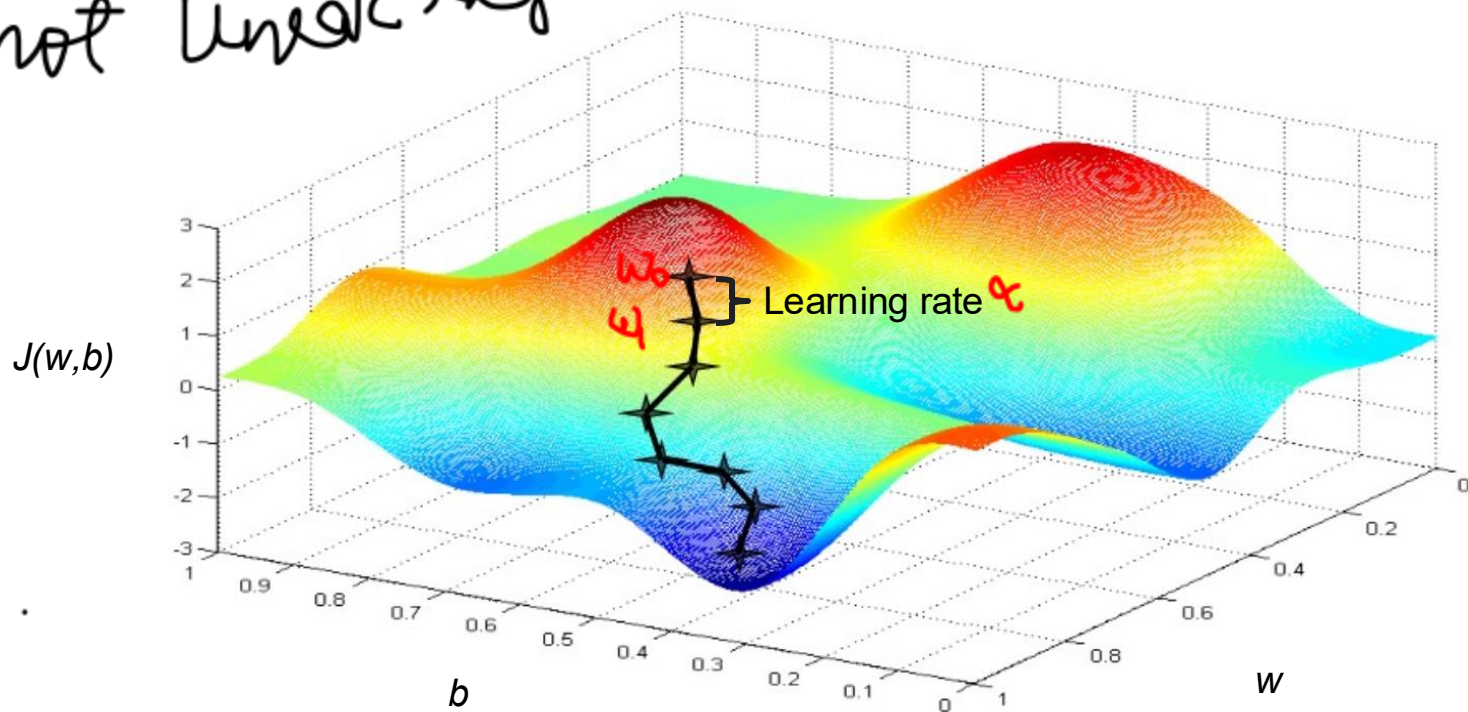
Gradient Descent- Learning Rate

- ▶ If α is too small, the rate of convergence to the minimum will be quite slow,
- ▶ If α is large, the successive iterates may oscillate wildly and you may never get to the minimum.



Gradient Descent

not linear regression



ANY QUESTIONS?